Parkinson's Disease Detection

Multimodal detection using MRI & DAT scans

Presented by

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Problem

Parkinson’s disease is a neurodegenerative disorder affecting basal ganglia, causing motor dysfunction. MRI helps analyze structural brain changes for diagnosis.

Objective

Develop an automated system using MRI and deep learning to detect Parkinson’s disease.

Contribution

Proposed hybrid framework combining ensemble deep learning, Elliptical Deviation Score, and Grad-CAM to improve reliability, interpretability, and diagnostic confidence in Parkinson’s detection.

Background & Existing Work

WHAT PRIOR WORK COVERS

- Deep learning models like CNNs and hybrid architectures are widely used for Parkinson's detection from MRI images.
- Studies analyze structural brain regions using segmentation and feature extraction techniques for disease identification.

GAPS IN EXISTING SYSTEMS

- Models often overfit and fail to generalize well on new data, reducing real-world reliability.
- Lack of structural validation; predictions rely only on learned features without confirming anatomical changes.

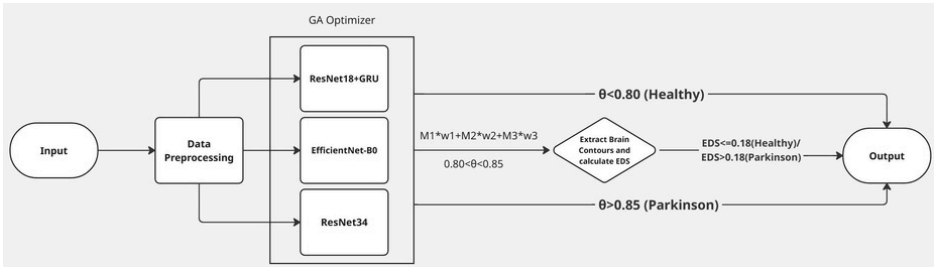
WHY THIS PROJECT IS NEEDED

- MRI changes in Parkinson's are subtle and often not clearly visible, making accurate diagnosis challenging.
- Manual analysis is time-consuming and inconsistent, requiring automated and reliable detection systems for clinical support.

HOW THIS PROJECT SOLVES IT

- Introduces Elliptical Deviation Score to quantify structural distortion in brain regions, adding anatomical validation.
- Combines deep learning prediction with EDS verification, improving reliability and reducing false positives in diagnosis.

Proposed System & Architecture



Working of the system

- MRI image is preprocessed and passed through ensemble deep learning models to generate Parkinson's disease probability score.
- Brain region is extracted and ellipse fitting is applied to compute Elliptical Deviation Score for structural validation.
- Final prediction is made by combining model confidence and EDS, with Grad-CAM highlighting important regions for explainability.

TOOLS USED

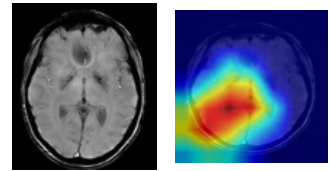
- Python
- PyTorch, Torchvision
- OpenCV, NumPy
- Flask
- Grad-CAM

Results & Analysis

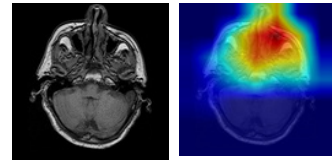
82%	1.0	0.89	0.5
Accuracy	PD Specificity	PD F1-Score	F1 Healthy

Approach	Loss	Acc	Prec	F1
ResNet18+GRU	BCE	0.79	0.77	0.87
ResNet18+GRU, EfficientNet-B0, ResNet34	BCE+pos_w	0.73	0.76	0.83
ResNet18+GRU, EfficientNet-B0, ResNet34	BCE	0.76	0.78	0.85
ResNet18+GRU, EfficientNet-B0, ResNet34	Focal loss	0.82	0.80	0.89

Positive Classification



Negative Classification



Conclusion & Future Work

WHAT WAS BUILT

- Developed ensemble deep learning system for Parkinson's detection using MRI images.
- Integrated Elliptical Deviation Score for structural validation and Grad-CAM for explainable predictions.
- Designed end-to-end pipeline from preprocessing to prediction with confidence-based decision fusion.

FUTURE IMPROVEMENTS

- Use accurate ROI detection (object detection) instead of contour-based brain extraction.
- Extend to 3D MRI analysis and validate on larger, multi-center clinical datasets.
- Improve EDS using region-specific analysis for more precise structural validation.

KEY NUMBERS

- High PD detection accuracy with ensemble model (strong confidence predictions above threshold 0.8).
- EDS reduces false positives and adds structural validation to improve reliability.
- Balanced performance with improved decision confidence using combined model prediction and EDS validation approach.

Thank You

Open to Q&A

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